

EXPERIMENT 2

8-Queens Problem

Question

Write a Python program to solve the 8-Queens problem.

Aim

To write a Python program to place **8 queens on an 8×8 chessboard** such that no two queens attack each other using the **Backtracking algorithm**.

Algorithm

1. Start with an empty chessboard.
2. Place a queen in the first row.
3. Check whether the position is safe.
4. If the position is safe, place the queen.
5. Move to the next row.
6. If no safe position is available, go back to the previous row.
7. Continue until all 8 queens are placed.
8. Display the solution.

Code

 python exp 2.2.py - C:/Users/royal/python exp 2.2.py (3.13.14)

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```
def is_safe(board, row, col):
    for i in range(row):
        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True

def solve(board, row):
    if row == 8:
        for i in range(8):
            for j in range(8):
                if board[i] == j:
                    print("Q", end=" ")
                else:
                    print("-", end=" ")
            print()
        return True

    for col in range(8):
        if is_safe(board, row, col):
            board[row] = col

            if solve(board, row + 1):
                return True

            board[row] = -1

    return False

board = [-1] * 8
solve(board, 0)
```

Output

```
===== RESTART: C:/Users/royal/python exp 2.2.py =====  
Q - - - - -  
- - - Q - - -  
- - - - - Q  
- - - - - Q - -  
- Q - - - - -  
- - - - - Q -  
Q - - - - -  
- - Q - - - -
```

Result

Thus, the **8-Queens problem** was **successfully solved using Backtracking**, and a valid arrangement of 8 queens was obtained.